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FACULDADE DE CIÊNCIAS E TECNOLOGIA
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Second written test of Experimental Methods in Computer Science

December 13th, 2024 Maximum time allowed: 1h:45m

Name:_____ **No.:**_____

Note: During this exam, students may consult their printed materials. The only electronic device allowed is a calculator, as long as it has no Internet connectivity. If you have any questions regarding the interpretation of the questions, indicate your understanding and respond accordingly.

1) Suppose you are testing the following hypothesis using a Z-test:

   H0: V2≤V1

   H1: V2>V1

The statistical test gives a Z-value of -2.35.

Under these circumstances, what is the P-value, and what conclusion should you reach? Justify your answer. (2.0/20)

2) If you have two samples with n < 30 and the data distribution is not normal, what are the consequences of using a two-sample T-test instead of a non-parametric method? Answer as thoroughly as possible. To make your answer clearer, it is suggested that you list the consequences one by one, rather than using a single sentence. (2.5/20)

3) A group of individuals representing typical users participates in the evaluation of an improved version of a sports brand's shopping App for iOS and Android phones. The evaluation consists of a set of five tasks (searching and purchasing various items) to be performed in the App, and the result is the total time each user spends completing the tasks. The new interface version (V2) is simpler than the previous version (V1), so it is expected that the time required to perform tasks with interface V2 will be shorter than with V1.

The experiment was organized as follows: the nine participants first used the original App (V1), performed the pre-prepared tasks, and their execution times were measured. The total time taken to complete the five tasks was calculated and recorded for each participant. Later, the same nine participants performed the same tasks using V2. Due to a technical issue, the task execution times of participant P7 could not be recorded. The following table shows the results:

	P1	P2	P3	P4	P5	P6	P7	P8	P9
V1	6.7	4.2	6.3	9.9	5.3	7.3	5.7	9.5	8.5
V2	4.8	7.4	5.1	4.7	6.9	5.1		8.4	6.9

Average	Std. Deviation
7.044	1.936
6.163	1.408

a) Based on these results, should V1 be replaced by V2? Consider a 95% confidence level. (6.5/20)

b) Suppose the experiment had the following differences:

- Participants have no prior knowledge or experience with either App version (V1 or V2).
- During the tasks, it was never indicated whether the participant was using V1 or V2.
- For each participant, it was randomly decided whether they performed tasks first in V1 and then in V2 or vice versa.

Would this new experimental setup be better than the one previously described? Justify your answer. Also, suggest what changes you would make to evaluate whether V1 should be replaced by V2 (only explain what you think should be changed; there is no need to repeat calculations). (2.5/20)

4) Your company intends to choose a lossless data compression algorithm and is considering four well-known algorithms: Gzip (Deflate), Bzip2 (Burrows–Wheeler), LZMA (Lempel–Ziv–Markov chain), and Zstandard.

Since compression rates depend on the nature of the data, the company decided to conduct an experiment with a significant sample of files from the company's data repository to determine whether there are significant differences in compression rates for the different algorithms. The dataset includes eight files and the experiment consists of measuring the compression rate of each algorithm for each file, in order to calculate the average compression rate of each algorithm and the standard deviation. The results are the following ones:

Algorithm	**Average compression rate**	**Std. deviation**
Gzip	5.76	1.06
Bzip2	6.07	0.98
LZMA	7.08	1.28
Zstandard	5.56	1.07

The compression rate values are presented in terms of the factor by which the original size must be divided to obtain the size after compression. For example, for Gzip, the average compression factor is 5.76. This means that when compressing a 100MB file, the compressed file will have a size of 100MB/5.76 = 17.36MB. Obviously, this is an estimate based on the average compression factor.

The speed of compression and decompression is not very important for the company, as the intended function is primarily for data archiving. Therefore, if the compression rates of the different algorithms are similar, the company will decide to purchase the commercial version of the cheapest algorithm (i.e., the decision is based solely on cost).

Based on a 95% confidence level, determine whether there are differences in the compression rates among the algorithms. Show your calculations, explain the process you used, and present your conclusion. (6,5/20)